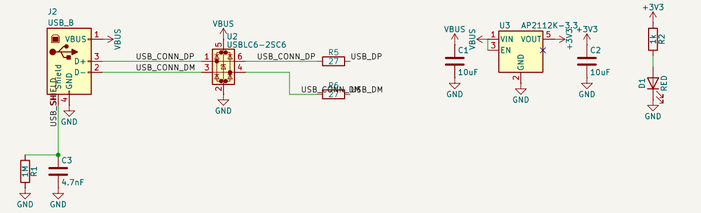
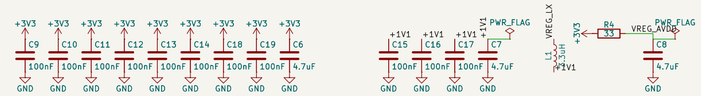
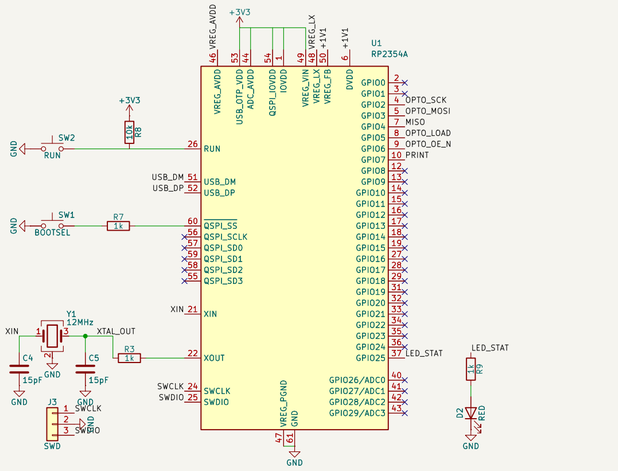
SOLARTRON 7075 DVM — ISOLATED USB INTERFACE (RP2354A)

USB SIDE (GND / +3V3 / VBUS)

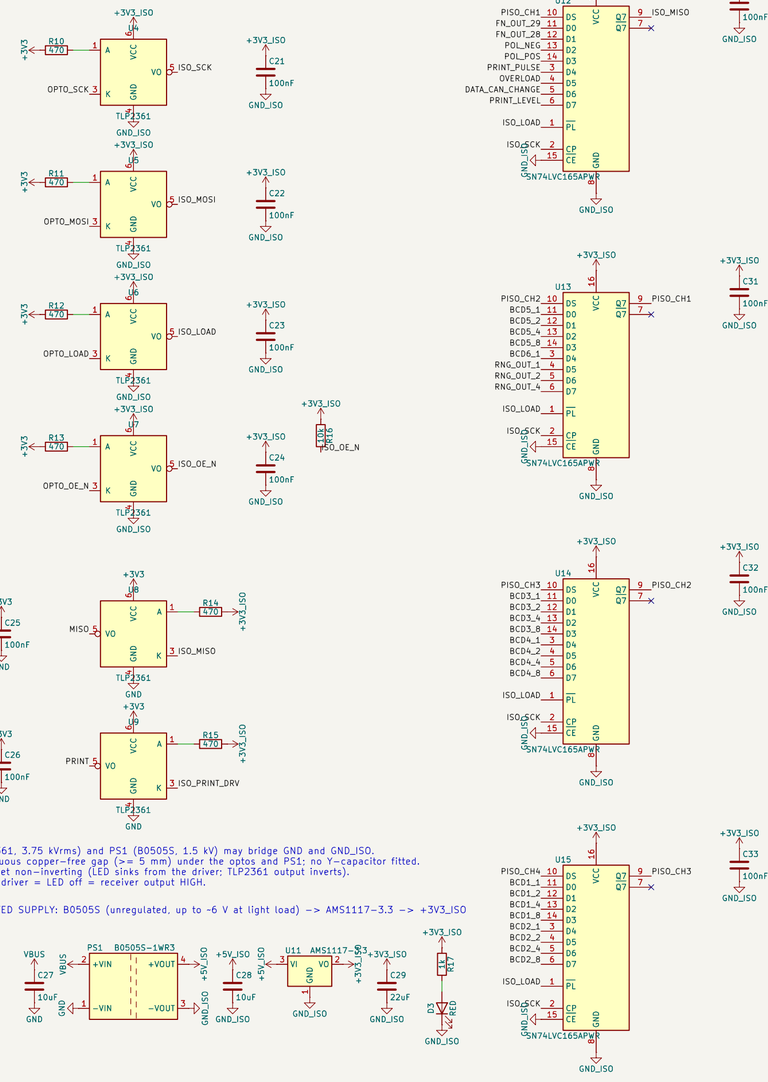


J2: TE 5787834-1 vertical USB-B on the TOP side (away from the meter).
RP2354A: internal 2 MB flash, QSPI data pins NC; BOOTSEL on QSPI_SS via R7.
L1 must be the polarity-marked AOTA-B201610S3R3 — place per RP2350 layout guide.





METER SIDE (GND_ISO = DD-50 pin 37 EARTH / logic 0, +3V3_ISO)



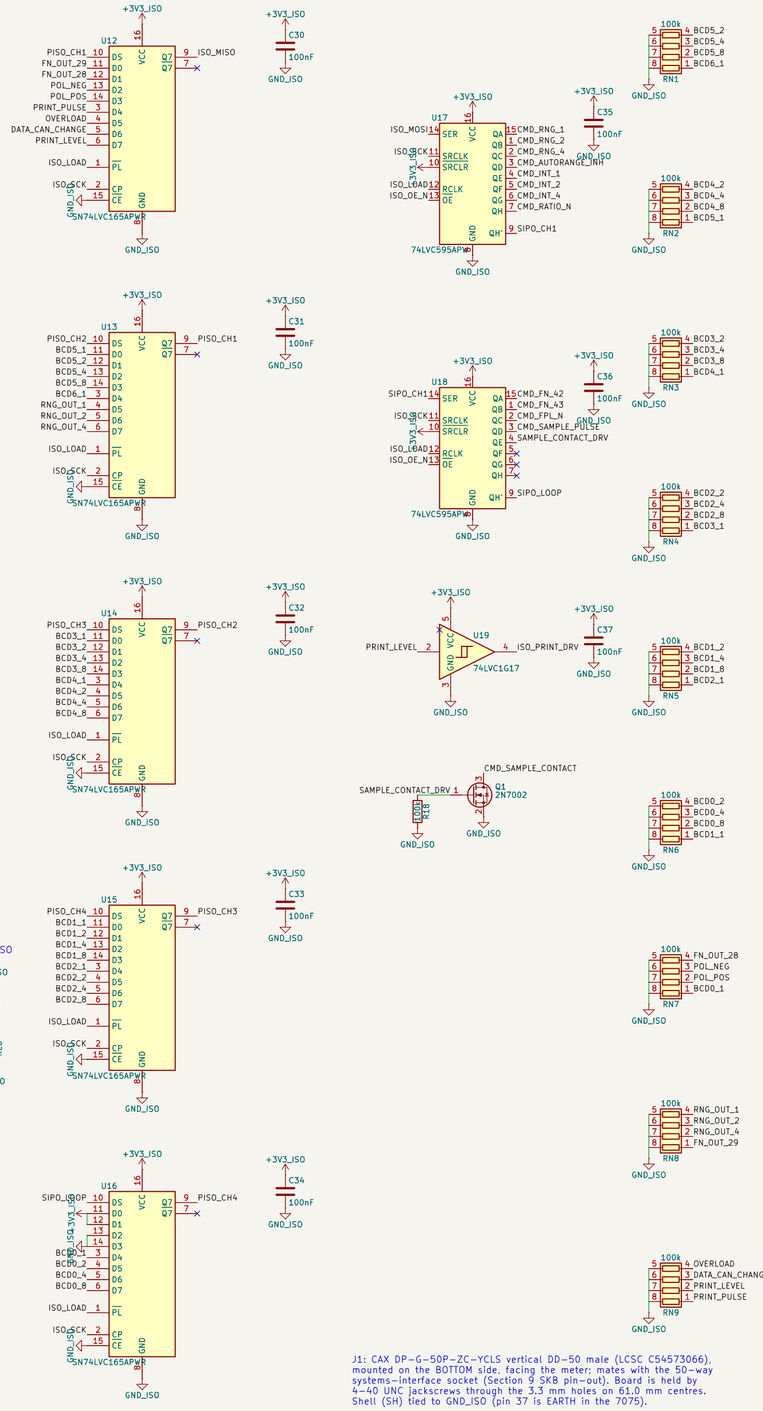
ISOLATION BARRIER
Only U4–U9 (TLP2361, 3.75 kVrms) and PS1 (B0505S, 1.5 kV) may bridge GND and GND_ISO.
PCB: keep a continuous copper-free gap (>= 5 mm) under the optos and PS1; no Y-capacitor fitted.
All opto links are net non-inverting (LED sinks from the driver; TLP2361 output inverts).
Unpowered / reset driver = LED off = receiver output HIGH.

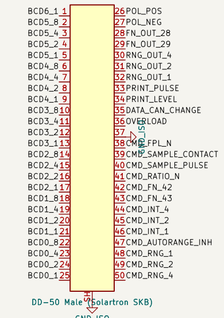
ISOLATED SUPPLY: B0505S (unregulated, up to ~6 V at light load) -> AMS1117-3.3 -> +3V3_ISO

SERIAL LINK ACROSS THE BARRIER (SPI mode 3, <= 1 MHz; PIO or SPIO)
GPIO2 SCK -> U4 -> ISO_SCK GPIO3 MOSI -> U5 -> ISO_MOSI GPIO5 LOAD -> U6 -> ISO_LOAD
GPIO6 OE_N -> U7 -> ISO_OE_N GPIO4 MISO <- U8 <- '165 chain GPIO7 PRINT <- U9 <- DVM pin 34 (IRQ)

ISO_LOAD low : all five '165s capture the 36 DVM outputs (asynchronous parallel load).
ISO_LOAD rising: both '595s latch the command word shifted in during the previous transfer.
Transfer = pulse LOAD, then 56 SCK clocks:
  MISO bits 1–40 : U12 D7..D0, U13 D7..D0, U14, U15, U16 D7..D0 (U16 D3..D0 = 0,0,1,1 signature)
  MISO bits 41–56: loop-back of the previous 16-bit command word (U18 QH' -> U16 DS)
  MOSI bits 41–56: new command word (first bit ends in U18 QH, last bit in U17 QA)
  Always resend the full command word — every LOAD edge re-latches the '595s.
ISO_OE_N is held high (R16 + opto idle) until firmware drives GPIO6 low after writing a valid word,
so with USB unplugged or the MCU in reset the DVM inputs float to their own defaults (front panel usable).

Meter-side logic is 74LVC at 3.3 V: inputs 5.5 V tolerant, V_IH 2.0 V (DVM high >= 2.4 V via 6k),
loff on all outputs — the DVM is never back-powered and pin 38 (FRONT PANEL LOCKOUT) is never pulled
low while this board is unpowered. RN1–RN9 (100k) keep the '165 inputs defined if the DVM is off.
SAMPLE: contact closure on pin 39 via Q1 (U18 QE high = closed), or >100 us pulse on pin 40 (U18 QD, 3.3 V).
NOTE: the Section 9 tables disagree on whether pin 42 or 43 is the AC-select bit — confirm on the instrument.





J1: CAX DP-G-50P-ZC-YCLS vertical DD-50 male (LCSC C54573066),
mounted on the BOTTOM side, facing the meter; mates with the 50-way
systems-interface socket (Section 9 SKB pin-out). Board is held by
4-40 UNC jackscrews through the 3.3 mm holes on 61.0 mm centres.
Shell (SH) tied to GND_ISO (pin 37 is EARTH in the 7075).

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